

Mini ERP + CRM Operations Portal

Full-Stack Technical Solution & Architecture Document

Position: Full Stack Developer | Location: Pune, Maharashtra | Role: Immediate Joining

1. Executive Summary & Overview

This document presents the comprehensive, production-ready solution for the Fundsroom Full Stack Developer Case Study. The project is an operational Mini ERP and CRM Portal engineered specifically for a wholesale and distribution enterprise. The system coordinates customer relationship lifecycles, catalog inventory with intelligent low-stock triggers, tamper-evident stock movement audit logs, and transactional sales delivery challans.

The implementation adheres strictly to the required modern web architecture: a high-performance React + TypeScript SPA on the client, an Express.js + TypeScript RESTful API on the backend, and a robust PostgreSQL database engine ensuring ACID guarantees.

2. Technical Stack Architecture

Tier / Component	Technology Chosen	Justification & Architectural Role
Frontend Client	React 18, TypeScript, Vite	Type safety, instant HMR, modular component structure, and sub-second bundle compilation.
UI & Styling	Custom Responsive CSS, Lucide Icons	Lightweight, ultra-responsive admin dashboard UI adhering to professional enterprise design guidelines.
Backend API	Node.js 22, Express.js, TypeScript	Scalable asynchronous RESTful service with strict request validation and centralized exception handling.
Database Tier	PostgreSQL (Neon Cloud & Docker)	ACID-compliant relational engine with row-level locking (FOR UPDATE) for zero-race-condition transactions.
Authentication	JWT (JSON Web Tokens) & Bcrypt	Stateless, cryptographic authentication with salted password hashing and role-based route middleware.
DevOps / Containers	Docker, Docker Compose, Nginx	Standardized containerization across database, backend API, and static frontend web server.
CI / CD Automation	GitHub Actions	Automated pipeline validating TypeScript compilation, linting, and production builds on push/PR.

3. Verified Test Login Credentials

All accounts are pre-seeded in the database with the uniform password:

Master Demo Password: Password@123

Role	Login Email	Password	Module Access & Permissions Scope
Admin	admin@fundsroom.local	Password@123	Full system access: CRM, Inventory, Stock Movements, Sales Challans, System Analytics.
Sales	sales@fundsroom.local	Password@123	Sales domain: Create/edit customers, log follow-ups, draft and confirm multi-item sales challans.
Warehouse	warehouse@fundsroom.local	Password@123	Inventory domain: Create/edit products, manage reorder alerts, record manual stock inward/outward.
Accounts	accounts@fundsroom.local	Password@123	Auditing domain: Read-only operational visibility, printable tax challans and transaction audit logs.

4. Core Modules Implementation Details

Module 1: Authentication & Role-Based Access Control (RBAC)

- Secured JWT authentication flow with bearer token header transmission.
- Express middleware 'allow(...roles)' rigorously enforces server-side role boundaries.
- Frontend shell dynamically renders role-specific badges, workspace designations, and contextual action controls.

Module 2: Customer CRM Management

- Fields tracked: Customer Name, Mobile, Email, Business Name, GST Number (Optional), Customer Type (Retail, Wholesale, Distributor), Address, Status (Lead, Active, Inactive), Follow-up Date, and Notes.
- Real-time search functionality querying customer names and business entities.
- Dedicated Customer Profile Drawer displaying contact records, registered GSTIN, and full history.
- Interactive Follow-up Log: Sales users log chronological interactions and dynamically reschedule the next action date.

Module 3: Products & Inventory Management

- Fields tracked: Product Name, SKU Code, Category, Unit Price (INR), Current On-Hand Stock, Minimum Stock Alert Threshold, and Warehouse Location.
- Dynamic low-stock indicators displaying red warning tags when on-hand quantity <= minimum threshold.
- In-place modal editing allowing warehouse managers to update price, stock levels, and alert rules.

Module 4: Stock Movement Audit Trail

- Immutable record logging: Product, SKU, Movement Type (IN / OUT), Quantity Changed, Business Reason, Authorized User, and Timestamp.
- Automatic background creation of 'OUT' movements upon challan confirmation, providing end-to-end trace integrity.

Module 5: Sales Challan Flow & Concurrency Safety

- Multi-Product Line Items: Sales users can dynamically add multiple product items to a single challan with live subtotal and inventory checks.
- Automatic Challan Number Generation in formatted sequence (e.g. CH-2026-XXXXXX).
- Draft vs. Confirmed status: Drafts reserve no inventory; Confirmed status atomically decrements inventory.
- Historical Pricing Snapshot: Challan line items preserve product name, SKU, and unit price in `challan_items` to guarantee invoice immutability.
- Row-Level Locking (SELECT ... FOR UPDATE) prevents race conditions and overselling.
- Zero Negative Stock Enforcement: If available inventory is insufficient, the transaction rolls back with HTTP 409 Conflict.

5. Bonus Points Implementation

- **1. Containerization:** Docker Setup: Complete `docker-compose.yml` orchestrating PostgreSQL, multi-stage Node backend, and Nginx frontend.
- **2. Automated CI/CD:** GitHub Actions CI: Fully automated `.github/workflows/ci.yml` pipeline for continuous integration and typechecking.
- **3. PDF Export:** Invoice & Delivery Challan PDF Export: One-click print/PDF layout with company headers, customer GST, itemized totals, and signatures.

6. Step-by-Step Local Execution Guide

The project is 100% verified and operational on the development workstation.

Step 1: Open VS Code terminal in the project directory (fundsroom project).

Step 2: Start both client and server concurrently by executing:

`pnpm dev` (or `npm run dev`)

Step 3: Open your browser and navigate to: `http://localhost:5173`

Step 4: Log in using any of the 4 test accounts listed in Section 3.

7. Screen Recording & Submission Checklist

As mandated in the assessment instructions, ensure the following steps are performed:

- **1. Screen Recording:** Start your screen recorder (OBS Studio, Windows Game Bar [Win + G], or Loom).
- **2. CRM Flow:** Demonstrate Admin login, overview metrics, and customer creation with GST number.
- **3. Follow-up Flow:** Demonstrate customer detail view and logging a follow-up note with new date.
- **4. Inventory Flow:** Demonstrate product catalog, low stock indicator, and editing a product.
- **5. Challan Flow:** Create a multi-product Sales Challan as Draft, view its details, and confirm it.
- **6. Stock Verification:** Demonstrate that stock was decremented and an OUT movement was created in the Audit log.

- **7. PDF Export:** Click 'Print / Export PDF' to demonstrate the delivery note invoice.
- **8. Drive Upload:** Upload recording to Google Drive (set permission to 'Anyone with the link can view').
- **9. Form Submission:** Submit the Google Form before 6:00 PM tomorrow with PDF solution & Drive link.

8. Immediate Joining Confirmation

I confirm my availability for immediate joining at Fundsroom in Pune, Maharashtra. I am fully comfortable with the Pune work location and look forward to contributing to Fundsroom's technology roadmap.